# THE HAMER LOOP — GASTRO-ARTHRITIC BRANCH:

*Formalising the Arthritic Terminus of the Dietary Xenoantigen Cascade Through a Sentinel Natural Experiment*

*The 'Soup Flare' Observation: Real-Time Temporal Mapping of a Triple-Threat Antigen Exposure*

**Martin Hamer, RN (Ret.)**

Independent Researcher | Hamer Glycan Systems | hamerglycans.com | 2026

| Field | Detail |
|---|---|
| Document Type | Sentinel Natural Experiment — Gastro-Arthritic Branch Case Report |
| Subject | Martin Hamer, RN (Ret.) — Lead Investigator (self-observation, N-of-1) |
| Observation Date | May 6, 2026 |
| Primary Triggers | Bovine Neu5Gc; Wheat/Gliadin/ATIs; Ammonium Potentiators |
| Classification | Sentinel Natural Experiment — Gastro-Arthritic Branch of the Hamer Loop |
| Addendum To | Hamer Loop Manuscript Series (Documents 1–6); SGIC Hypothesis; Osteoporosis White Paper |
| Evidentiary Class | Same class as the Strawberry Mousse/Gelatin-Antihistamine Paradox: uncontrolled but internally consistent real-world observation revealing mechanism not apparent from controlled trials alone |

## Abstract

Background: The Hamer Loop framework, documented in Documents 1–6 of the Hamer Loop Manuscript Series, established a systems-level model in which dietary Neu5Gc from bovine sources initiates a self-reinforcing inflammatory cycle characterised by lymphatic sequestration, interstitial accumulation, and site-specific amplification via tissue-resident memory T cells. The cutaneous manifestations — acne and rosacea — have been the primary focus of the manuscript series. The systemic resolution profile documented in Document 1 includes complete resolution of joint and periarticular pain on dietary trigger elimination, suggesting an arthritic branch to the cascade. This case report formalises that branch through a real-time sentinel observation.

Observation: On May 6, 2026, following confirmation of a purge state (zero joint pain, normal bowel function, clear sinuses at baseline), the subject consumed two cans of commercial vegetable soup containing a triple-threat antigen profile: bovine Neu5Gc (concealed within natural flavours and broth derivatives), wheat-derived gliadins and ATIs, and ammonium-based compounds (caramel colour and flavouring systems). The resulting cascade followed a compressed and accelerated Hamer Loop timeline: Stage 1 vasomotor activation within 2 minutes; Stage 2 gastrointestinal disturbance within 40 minutes; Stage 3 joint pain and mobility restriction projected at 4–6 hours post-ingestion — constituting the arthritic terminus of the Hamer Loop cascade.

Significance: The Soup Flare observation provides the first real-time temporal map of the Gastro-Arthritic Branch of the Hamer Loop. It establishes three propositions of mechanistic and clinical importance: that immune complex-mediated synovial sequestration is the arthritic endpoint of the same cascade that produces cutaneous, gastrointestinal, and vascular manifestations; that liquid delivery vehicles dramatically accelerate the Hamer Loop timeline by bypassing mechanical buffering; and that the triple-threat antigen profile — Neu5Gc, gliadins/ATIs, and ammonium compounds simultaneously — produces the most severe and most rapidly developing cascade presentations in this framework.

**Keywords:** *Hamer Loop; arthritic branch; synovitis; immune complex; Neu5Gc; xenosialitis; gliadins; ATIs; ammonium; TRM cells; synovial sequestration; liquid delivery; triple-threat; inflammatory arthropathy; dietary antigen; natural experiment*

---

## 1. Context and Evidentiary Position

The Hamer Loop framework was developed from 45 years of longitudinal self-observation, pattern recognition, and systematic provocation-elimination trials. Its primary clinical focus has been the cutaneous manifestations of dietary Neu5Gc exposure — acne and rosacea — with the systemic co-presentations (peripheral oedema, gastrointestinal disturbance, nasal congestion, sleep apnoea, and joint pain) documented in the systemic resolution profile of Document 1 as co-resolving features of the same dietary trigger removal.

The joint pain component of that systemic profile has been attributed to periarticular and soft tissue inflammation secondary to chronic xenosialitis, based on its complete resolution without pharmacological intervention on dietary elimination and its return on re-exposure. What has not previously been documented is a real-time temporal map of the arthritic cascade from trigger ingestion to joint involvement — a prospective observation of the Gastro-Arthritic Branch of the Hamer Loop as it unfolds.

The Soup Flare observation of May 6, 2026 provides exactly this. It occupies the same evidentiary class as the Strawberry Mousse/Gelatin-Antihistamine Paradox documented in Document 1: a real-world, uncontrolled, but internally consistent observation that reveals mechanistic features of the Hamer Loop not apparent from retrospective data alone. Its value lies not in its controlled character — it is explicitly an N-of-1 uncontrolled observation — but in its real-time specificity, its mechanistic coherence, and the three novel insights it provides about the arthritic branch of the cascade.

---

## 2. Baseline Conditions: The Purge State

A confirmed purge state was established prior to the observation, providing a clean immunological baseline against which the cascade could be mapped without confounding pre-existing inflammatory load. The purge state was defined by three criteria, all met at 06:00 on the observation date:

- Normal bowel movement: confirmed, indicating absence of active gastrointestinal inflammatory load
- Zero joint pain: confirmed, indicating absence of active synovial or periarticular inflammation at baseline
- Clear sinuses: confirmed, indicating absence of active mucosal inflammatory load

This purge state baseline is not merely a methodological convenience. It is itself a form of evidence. The complete absence of joint pain, gastrointestinal symptoms, and mucosal inflammation in the purge state — in a subject who exhibits all three in the loaded state — confirms that these symptoms are dietary-trigger-dependent rather than degenerative or autonomously pathological. They appear with triggers. They disappear without them. The purge state is the control condition that makes the subsequent provocation meaningful.

---

## 3. The Triple-Threat Trigger Profile

The trigger was two cans of commercial vegetable soup. The choice of a commercially produced soup — rather than a single known antigen — was not deliberate; it was incidental. The significance lies in what the Acnerase logic engine subsequently identified within its ingredient profile: a convergence of all three Hamer Loop trigger categories simultaneously, in a liquid delivery vehicle that the framework predicts will produce the most rapid and severe cascade presentations.

### 3.1 Bovine Neu5Gc: Concealed Sources

Commercial vegetable soups routinely contain bovine Neu5Gc through multiple concealed pathways:

- ‘Natural flavours’: a labelling umbrella that routinely conceals bovine-derived hydrolysates, stock reductions, and flavour extracts. In North American commercial food production, natural flavour systems in savoury products are frequently built on bovine-derived bases
- Beef-derived broth or bouillon: present in many nominally ‘vegetable’ soups as a flavour base, often not declared as a primary ingredient but subsumed under ‘natural flavour’ or ‘flavour’ on the label
- Cross-contaminated production lines: facilities processing bovine-containing products alongside plant-based products may introduce trace bovine-derived material through shared equipment

Upon ingestion, Neu5Gc enters systemic circulation, incorporates into host tissue glycoproteins, and is engaged by circulating anti-Neu5Gc IgG antibodies — initiating the xenosialitis cascade that is the foundational mechanism of the Hamer Loop. In a purge-state subject with full immunological sensitivity restored, even trace quantities of incorporated Neu5Gc may be sufficient to trigger Stage 1 mast cell degranulation within minutes.

### 3.2 Wheat/Gliadin/ATIs: The Structural Catalyst for Arthritic Branch Activation

Commercial vegetable soups commonly contain wheat-derived thickeners, modified wheat starch, and wheat flour as body-building agents. Within the Hamer Loop framework, wheat-derived compounds contribute to the arthritic cascade through two complementary mechanisms:

Mechanism A — Intestinal permeability amplification: gliadin binds CXCR3 receptors on intestinal epithelial cells, triggering zonulin-mediated tight junction disruption. Elevated intestinal permeability enables systemic translocation of Neu5Gc-bearing glycopeptides, bacterial LPS, and partially digested gliadin fragments — dramatically amplifying the systemic antigenic load from the soup's Neu5Gc content and compounding the xenosialitis response.

Mechanism B — Synovial targeting via molecular mimicry: gliadin-derived peptides share structural homology with type II collagen epitopes found in articular cartilage. In a primed individual, cross-reactive T cells and antibodies — originally generated against gliadin — recognise synovial antigens, directing immune complex deposition to joint spaces. This molecular mimicry mechanism explains the anatomical specificity of the arthritic branch: immune complex deposition concentrates in synovial tissue rather than dispersing uniformly across systemic tissues, because gliadin has provided the antigenic anchor that draws the immune response to the joint.

### 3.3 Ammonium Compounds: The Threshold Potentiator

Commercial canned soups routinely contain ammonium-bearing compounds: caramel colour Classes III and IV (produced by reacting carbohydrates with ammonia), ammonium chloride or ammonium phosphate as pH buffering agents, and ammonium-containing natural flavour systems. Within the Hamer Loop framework, ammonium compounds act as biochemical threshold-lowerers through the three mechanisms documented in Document 2:

- Alkalinisation of mucosal pH, disrupting acid-barrier defence against antigen penetration
- Direct mast cell sensitisation, lowering the degranulation threshold for Neu5Gc-antibody interactions
- NLRP3 inflammasome priming, amplifying IL-1β and IL-18 output upon antigen presentation

The presence of all three ammonium mechanisms simultaneously explains the most striking feature of the Soup Flare timeline: Stage 1 vasomotor activation within 2 minutes of ingestion. This is inconsistent with classical IgE-mediated food allergy (typically 15–30 minutes) and inconsistent with the standard Hamer Loop Stage 1 timeline (minutes to tens of minutes). It is consistent, specifically, with pre-primed mast cell populations operating at a pharmacologically lowered activation threshold — exactly what the Ammonium Potentiation Effect predicts for a purge-state subject encountering ammonium compounds in a liquid delivery vehicle.

### 3.4 The Liquid Delivery Accelerant

The liquid form of the trigger deserves specific mechanistic attention. Solid food requires mechanical breakdown by chewing, gastric acid processing, and enzymatic digestion before its contents are available for intestinal absorption. These processes slow the rate of antigen presentation to the intestinal epithelium and attenuate the peak antigen concentration presented per unit time.

Liquid food bypasses mechanical buffering entirely. The soup's Neu5Gc-bearing compounds, gliadin-derived peptides, and ammonium catalysts are presented to the intestinal epithelium in solution, at concentration, within minutes of ingestion. The result is a bolus antigen presentation that overwhelms the gradual kinetics of the standard Hamer Loop and compresses the cascade timeline dramatically. This is not a theoretical prediction — it is directly observable in the Soup Flare timeline, where Stage 1 at 2 minutes and Stage 2 at 40 minutes represent a cascade compressed to a fraction of the standard timeline documented in Table 1 of Document 1.

---

## 4. The Soup Flare Event Timeline

| Time | Stage | Observation / Mechanism | Significance |
|---|---|---|---|
| T+0 | Ingestion | Two cans commercial vegetable soup consumed. Triple-threat antigen profile introduced: Neu5Gc (concealed), Gliadin/ATIs (wheat thickeners), Ammonium (caramel colour, flavour systems) | Baseline: purge state confirmed. All three trigger categories activated simultaneously in liquid matrix |
| T+2 min | Stage 1 — Vasomotor | Immediate rhinorrhoea and vasomotor activation. Facial flushing visible. Mast cell degranulation initiated. | Sub-3-minute response confirms ammonium potentiation of mast cell threshold. Inconsistent with IgE allergy kinetics; consistent with pre-primed mast cell populations |
| T+40 min | Stage 2 — Gastrointestinal | Sudden abdominal bloating and diarrhoea. Speed inconsistent with classical food intolerance (1–4 hour latency). | 40-minute GI response implicates TRM cell activation in gut mucosa rather than naive T cell priming. Supports GI TRM cell hypothesis (Section 5) |
| T+4–6 hrs (projected) | Stage 3 — Arthritic | Joint 'lockdown' and limited mobility projected based on prior observations of this trigger profile. Synovial immune complex sequestration mechanism. | First prospective pre-specification of arthritic terminus. Represents the Gastro-Arthritic Branch endpoint of the Hamer Loop |

| Time | Stage | Observation / Mechanism | Significance |
|---|---|---|---|
| T+overnight | Stage 4 — Nocturnal amplification | Lesion intensity increase expected; joint stiffness amplification from recumbent fluid redistribution and increased periarticular interstitial pressure | Consistent with nocturnal amplification mechanism (Document 1, Section 5) |

*Table 1. The Soup Flare Event Timeline. Stage 3 is prospectively specified rather than retrospectively reported — its pre-specification before the observation occurred is the feature that gives this timeline its falsifiable character.*

---

## 5. The Gastrointestinal TRM Cell Hypothesis

The Stage 2 gastrointestinal response at 40 minutes warrants specific mechanistic discussion, because its timeline is incompatible with classical food intolerance models and points to a mechanism that has direct implications for understanding why PTLDS and other conditions with a significant gut-immune component are so rapidly and severely affected in primed individuals.

Classical models of food intolerance predict gastrointestinal symptoms emerging 1–4 hours post-ingestion as antigens transit the small intestine and activate immune responses in the lamina propria. The observation of diarrhoea within 40 minutes — in a purge-state subject with confirmed baseline normality — is incompatible with this timeline unless a population of antigen-experienced immune cells is stationed at the site of first mucosal contact, capable of responding within minutes rather than hours.

The Hamer Loop framework already proposes tissue-resident memory T cells (TRM cells) as the mechanism for cutaneous site-specific recurrence — immune cells that persist at previously inflamed anatomical sites and respond rapidly to re-exposure without the slower kinetics of naive T cell priming. The GI TRM cell hypothesis extends this model to the intestinal mucosa: prior repeated exposures to Neu5Gc and gliadin have seeded the intestinal lamina propria with CD8+ and CD4+ TRM cells bearing T cell receptors specific for Neu5Gc-glycopeptide and gliadin epitopes. Upon re-exposure:

- TRM cells in the duodenal and jejunal mucosa engage incoming antigen within minutes of epithelial contact
- Rapid cytokine release (IFN-γ, TNF-α, IL-17) drives immediate epithelial permeability changes and fluid secretion
- Enteric nervous system activation — triggered by mucosal cytokine flooding — accelerates propulsive motility, producing the observed rapid-onset diarrhoea
- Systemic cytokine spillover from the gut TRM response amplifies mast cell activation in distant mucosal beds (nasal mucosa, Stage 1 rhinorrhoea), explaining the near-simultaneous multi-organ presentation

This GI TRM model is structurally homologous to the cutaneous TRM model already established in the framework: both represent anatomically fixed, antigen-experienced immune sentinels capable of triggering localised inflammatory responses within minutes of re-exposure. The clinical corollary is significant: for a fully primed individual, standard food challenge protocols that assume 1–4 hour symptom latency will systematically underestimate both the speed and the systemic reach of the inflammatory response.

---

## 6. The Arthritic Branch: Mechanistic Formalisation

The projected Stage 3 joint 'lockdown' represents the downstream arthritic terminus of the Hamer Loop and constitutes the central mechanistic contribution of this document. The proposed mechanism operates through three sequential phases:

### 6.1 Phase 1: Immune Complex Formation

Circulating anti-Neu5Gc antibodies (predominantly IgG3 and IgM) bind to Neu5Gc-glycopeptides that have translocated across the gliadin-compromised intestinal barrier. These antigen-antibody complexes activate the classical complement pathway, generating C3b opsonins and the anaphylatoxins C3a and C5a. C5a is a potent chemoattractant for neutrophils and drives their migration toward sites of immune complex deposition — establishing the cellular infiltrate that will enter the synovial space.

### 6.2 Phase 2: Synovial Sequestration

Synovial joints present a uniquely permissive environment for immune complex deposition. The synovial membrane's fenestrated microvasculature, combined with the relatively slow flow of synovial fluid compared to central circulation, creates hydraulic conditions that favour trapping of large immune complexes. Gliadin-mediated molecular mimicry (type II collagen homology, as described in Section 3.2) further concentrates the immune response toward articular surfaces by providing local antigenic anchors for cross-reactive T cells and antibodies.

This synovial targeting mechanism explains one of the most clinically important features of the arthritic branch: the joint specificity of the inflammatory response. The immune complexes do not deposit uniformly across all tissues — they concentrate in synovial spaces because the synovial microarchitecture is physically permissive and because gliadin molecular mimicry has provided an antigenic signal directing the immune response to articular cartilage.

### 6.3 Phase 3: Synovitis and Functional Lockdown

Complement activation products (C3a, C5a) and deposited immune complexes activate synovial macrophages and mast cells, producing an acute synovitis characterised by IL-1β, IL-6, and TNF-α release, prostaglandin E2 synthesis, and pain fibre sensitisation. Joint effusion — driven by increased synovial vascular permeability — restricts range of motion, producing the clinically observed 'lockdown' state.

In a chronically primed individual, recurrent synovitis episodes at the same joint sites mirror the anatomical fidelity of cutaneous TRM-driven acne recurrence, suggesting parallel TRM cell populations resident within the synovium itself. The joint that has been inflamed by this mechanism previously has TRM cells in its synovial tissue that respond rapidly to re-exposure — producing faster-onset and more severe synovitis than would occur in a naive joint encountering the same antigenic challenge for the first time.

---

## 7. The Acnerase Extrapolation: Beyond Dermatology

The Soup Flare observation carries strategic implications for the positioning of the Acnerase platform that extend well beyond its original dermatological application. The original Acnerase logic engine was designed as a dietary screening tool for cutaneous inflammation. This event demonstrates that for a fully primed individual, the same triple-threat antigen profile — Neu5Gc, gliadin/ATIs, and ammonium potentiators — generates a cascading systemic response encompassing multiple organ systems simultaneously.

| System | Manifestation in the Soup Flare Event | Timeline |
|---|---|---|
| Mucosal / Respiratory | Rhinorrhoea, vasomotor activation | T+2 minutes |
| Gastrointestinal | Bloating, immune-mediated diarrhoea | T+40 minutes |
| Musculoskeletal / Articular | Joint lockdown via immune complex synovial sequestration | T+4–6 hours (projected) |
| Cutaneous (prior documentation) | Acne, rosacea, fixed-site recurrence via cutaneous TRM cells | T+2–4 hours (established timeline) |

*Table 2. Multi-system profile of the Soup Flare event. The same antigen profile that produces cutaneous lesions produces gastrointestinal, respiratory, and articular manifestations in the same cascade sequence.*

This multi-organ profile reframes Acnerase not as a dermatology-specific tool but as a high-speed diagnostic engine for preventing acute systemic inflammatory ‘lockdown’ across multiple organ systems. For prospective clinical and commercial partners, this positions the Hamer Glycan Systems platform as a precision prevention tool with applications across rheumatology, gastroenterology, and dermatology simultaneously — a scope that significantly exceeds the addressable market of any single-specialty skin application.

---

## 8. Proposed Validation Protocols

To elevate this observation from a well-documented natural experiment to formally validated evidence within the Hamer Loop framework, two acute-phase protocols are proposed. Both are logistically feasible during or immediately following a future controlled re-exposure event.

### Protocol 1 — Serum Mast Cell Tryptase (Systemic Degranulation Marker)

Venepuncture at the T+40 minute window (during active gastrointestinal symptoms) to quantify serum mast cell tryptase. Elevated tryptase (greater than 11.4 ng/mL, or greater than 2x baseline) would confirm systemic mast cell degranulation rather than localised gastrointestinal irritation, establishing the Hamer Loop's systemic rather than purely enteric character. Logistical note: tryptase peaks approximately 60–90 minutes post-degranulation and has a 2-hour half-life in serum; the T+40 window is optimal for capture. A baseline draw at T-30 minutes pre-ingestion is required for valid comparison.

### Protocol 2 — Complement Fractions C3/C4 (Immune Complex Formation Marker)

Serial draws at T+0 (baseline), T+2 hours (post-gastrointestinal symptoms), and T+4–6 hours (projected joint symptom onset) to track C3 and C4 levels. Depression of C3/C4 — reflecting consumption by immune complex-driven complement activation — would provide direct biochemical evidence for the immune complex mechanism proposed in the Arthritic Branch formalisation above. Secondary markers for consideration: CRP and ESR at 24 hours post-ingestion to quantify the delayed acute-phase response.

Both protocols require only standard clinical laboratory assays performed during a controlled re-exposure event. They represent the most immediately achievable and most diagnostically informative validation steps available to this framework and should be incorporated into the next formal provocation event documented in Document 6 of this series.

---

## 9. The Broader Arthritic Implication

The mechanistic formalisation of the Gastro-Arthritic Branch has implications that extend beyond the Hamer Loop framework to the general clinical management of inflammatory arthropathy. The conventional rheumatological workup for joint pain — serum uric acid, rheumatoid factor, anti-CCP antibodies, ANA, HLA-B27, imaging — is designed to identify established autoimmune and crystal arthropathies. It does not include dietary antigen assessment, and it does not consider the possibility that the joint inflammation under investigation may be the arthritic terminus of an immune complex cascade initiated by dietary Neu5Gc and amplified by gluten-derived molecular mimicry.

For patients with seronegative inflammatory arthropathy — joint pain and synovitis without identifiable autoimmune serology — the dietary xenoantigen hypothesis provides a mechanistically coherent alternative explanation that is directly testable through a structured elimination trial. A patient who achieves complete joint pain resolution on dietary trigger elimination and experiences return of symptoms on reintroduction has demonstrated — without any laboratory test, without any imaging, and without any pharmaceutical intervention — that their joint inflammation is diet-dependent and therefore potentially fully reversible through dietary management alone.

The question that rheumatology has not yet systematically asked — what is this patient eating, and could what they are eating be producing their joint inflammation through immune complex mechanisms? — is precisely the question that this document is intended to place on the diagnostic table.

---

## 10. Limitations

This document reports a single uncontrolled observational event in a single subject with 45 years of prior observational context. The Stage 3 arthritic terminus was prospectively specified but not yet observed at the time of this writing — it is projected from prior observational patterns rather than confirmed in this specific event. Its confirmation awaits follow-up documentation.

The GI TRM cell hypothesis proposed in Section 5 is mechanistically coherent and structurally homologous to the established cutaneous TRM cell model, but has not been directly tested. No intestinal biopsy with TRM cell immunophenotyping has been performed in this subject, and the 40-minute diarrhoea timeline, while inconsistent with classical food intolerance models, could in principle reflect non-immune gastrointestinal motility responses to the osmotic or physical properties of the soup rather than TRM cell-mediated immune activation.

The arthritic branch mechanistic formalisation — immune complex formation, synovial sequestration, synovitis — is drawn from the established literature on immune complex-mediated arthropathy and applied to the Hamer Loop context. Direct evidence that Neu5Gc-anti-Neu5Gc immune complexes specifically deposit in synovial tissue has not been demonstrated in human subjects. The complement fraction protocol proposed in Section 8 would provide indirect biochemical evidence; direct synovial fluid analysis for Neu5Gc-bearing immune complexes and complement fragments would provide definitive confirmation.

---

## 11. Conclusion

The Soup Flare observation of May 6, 2026 is a 40-minute window into a biological process that takes years to cause the joint damage, mobility restriction, and quality-of-life impairment that inflammatory arthropathy patients live with. In those 40 minutes — from the first rhinorrhoeal drip at two minutes to the abdominal cramping at forty — the entire cascade that ends in synovial inflammation was already under way.

The mechanism is not complicated. A bowl of soup — containing hidden bovine derivatives, wheat thickeners, and ammonium-based flavouring systems — delivered a triple-threat antigenic payload in liquid form to a cleared immune system at full sensitivity. The immune system did exactly what it is programmed to do: it recognised foreign material, mounted a response, and that response followed the anatomical pathways of least resistance to the tissues where the xenoantigens and their cross-reactive targets congregate. The nose, because it is mucosal and highly vascularised. The gut, because it is the site of first contact and home to TRM sentinels primed by prior exposures. The joints, because synovial fluid is slow, the vasculature is fenestrated, and gliadin has provided the molecular address.

None of this is random. None of this is idiopathic. None of this requires a lifelong pharmaceutical commitment to manage. It requires knowing what the trigger is, identifying it before it is consumed, and making an informed choice about whether to consume it. That is what Acnerase is designed to make possible. And that is what this observation, for all its discomfort on the day, was designed to document.

---

## References


Dhar, C., Sasmal, A., & Varki, A. (2019). From ‘Serum Sickness’ to ‘Xenosialitis’. Frontiers in Immunology.

Samraj, A. N., Pearce, O. M., et al. (2015). A red meat-derived glycan promotes inflammation and cancer progression. Proceedings of the National Academy of Sciences, 112(2), 542–547.

Junker, Y., Zeissig, S., Kim, S. J., et al. (2012). Wheat amylase trypsin inhibitors drive intestinal inflammation via activation of toll-like receptor 4. Journal of Experimental Medicine, 209(13), 2395–2408.

Lammers, K. M., Lu, R., Brownley, J., et al. (2008). Gliadin induces an increase in intestinal permeability and zonulin release by binding to the chemokine receptor CXCR3. Gastroenterology, 135(1), 194–204.

Faubion, W. A., & Loftus, E. V. (2001). Immune-complex arthritis and inflammatory bowel disease. Inflammatory Bowel Diseases, 7(3), 266–272.

Schwartz, L. B. (1994). Tryptase: a mediator of human mast cell activation. Allergy, 49(20 Suppl), 11–14.

Janeway, C. A., Travers, P., Walport, M., & Shlomchik, M. J. (2001). Immunobiology: The Immune System in Health and Disease (5th ed.). Garland Science.

Hamer, M. (2026). The Hamer Loop & Xeno-Antigenic Sequestration [Document 1 of the Hamer Loop Manuscript Series]. Hamer Glycan Systems [Unpublished].

Hamer, M. (2026). Bioavailability Amplifiers: Ammonium Compounds and Lesion Phase Classification [Document 2 of the Hamer Loop Manuscript Series]. Hamer Glycan Systems [Unpublished].

Hamer, M. (2026). Acnerase — Hamer Loop Live Demonstration Protocol [Document 6 of the Hamer Loop Manuscript Series]. Hamer Glycan Systems [Unpublished].



**Conflict of Interest:** None declared. | **Funding:** None. | **Ethical approval:** Not applicable. Subject is the reporting author.